FLEET TEAM DIGEST

737MAX-FTD-34-18008

Issue Title : FMC Hardware Model C1 In-Air Restarts Due to Static Random Access Memory (SRAM) Multiple Bit Errors

Airplane Model	ATA	Minor Model(s)
737MAX	3461-00	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
737NG	9E991	

Originated Date	Last Revised Date	Created On
07/16/2018	09/10/2026	07/09/2026
Estimated Completion Date	**Next Update Date**	
01/30/2027	12/05/2026	

Revision Description

Updated the following section(s):

- Final Action
- Milestones
- Part Information

Applicability

This FTD is applicable to all 737 MAX Airplanes equipped with "C1" FMC Hardware (P/N 10-62225-004, GE Model No.: 2907C1, GE P/N 176200-01-01).

Description

Review of in-service FMC Fault Logs for the "C1" FMC has found a significant increase of FMC restarts due to a particular fault code, 0x1009. The 0x1009 fault code indicates the FMC performed a cold restart in response to detection of a multiple bit error within the SRAM. A cold restart correlates to a clearing of all SRAM (loss of flight plan) and a longer restart time than a warm restart (typical response to a reset). A noncompliance has been filed for the C1 FMC (Boeing P/N 10-62225-004/GE P/N 176200-01-01). To restore compliance, Boeing will be certifying a new hardware part that includes different SRAM.

Background

The FMC hardware contains circuitry to detect single-bit errors and some multi-bit errors and correct single-bit errors within a 32-bit word; it does not have the ability to determine which bits are corrupted in the case of a multiple bit error. As a result, the FMC response to a multiple bit error is to clear the corrupted word at the affected address and perform a cold start to clear the entire SRAM. The entire SRAM is cleared as other data may depend upon the data at the corrupted address.

The investigation into these events found a significant increase of reports that correspond to the SRAM component change

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(SRAM P/N 173210-08 replaced by P/N 173210-11) made in 2012 due to component obsolescence. Moreover, GE and the SRAM component manufacturer confirmed these multiple bit errors (0x1009 fault codes) were primarily due to upsets within the SRAM component due to atmospheric radiation.

In 2018, a less-susceptible SRAM component was identified (SRAM P/N 173210-13) as a replacement for SRAM P/N 173210-11. This component was validated by Boeing/GE engineering and incorporated into production of the Model C1 FMCs starting at serial number 15356. Unfortunately, this component was later found to be responsible for a dual 0x1009 reset on a flight test. The resultant investigation found the root cause to be an internal modification to the component, inserted between its qualification and released into production. This resulted in a recall of all C1 FMCs with SRAM P/N 173210-13 for replacement with SRAM P/N 173210-11.

Boeing has not identified any 737 MAX airplanes delivered with SRAM P/N 173210-13 installed. All 737 MAX airplanes will be delivered with SRAM P/N 173210-11 installed until new SRAM parts are certified. The "C1" FMC with SRAM P/N 173210-08 or SRAM P/N 173210-11 is referred to as MOD 2 in other service bulletins. The "C1" FMC with SRAM P/N 173210-13 is referred to as MOD 4 in other service bulletins.

In the process of evaluating new SRAM components for a suitable replacement, GE Aviation performed atmosphere radiation testing on the current SRAM to establish a baseline. The test results show that the Cypress SRAM susceptibility to single and multi-bit upsets is higher than expected and that it does not meet the required probability for “Loss of LNAV during RNP operations with minima less than 0.3 nm”. As such, a non-compliance has been filed for the "C1" FMC. To restore compliance, Boeing will be certifying a new hardware part that includes a different SRAM. This item has been reviewed via the Boeing Safety process and was deemed "Not Safety".

Status

Field experience and manufacturer test data have revealed that the current 173210-11 SRAM used in the FMC (Cypress CY62148E, GE P/N 173210-11) is vulnerable to Single Event Effects (SEE) caused by normal atmospheric radiation and a small number of Multiple Bit Upset (MBU) within a single byte of SRAM. Cypress CY62148E, GE P/N 173210-11 SRAM part was found to be non-compliant, which resulted in the "C1" FMC being non-compliant.

A new SRAM (Microcircuit), made by ISSI, has been qualified; this new SRAM will be certified under a new FMC hardware part number to restore compliance.

New SRAM Vendor P/N: IS62C5128EL-45TLI

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New SRAM GE P/N: 200-003-776-001

New FMC HW P/N: 176200-02-01 (2907C2)

Interim Action

The only way to determine if an FMC restart is the result of a 0x1009 type fault is to provide FMC Fault Log data per Boeing AMM Task 34-61-00-810-801/802 as required.

A typical restart takes approximately 5 to 10 seconds, however with this type of error, the FMC is required to clear the SRAM memory of any corrupted data and may take 30-60 seconds.

As a means to support the certification of the forthcoming U14.1 FMC software, the Boeing team determined that an AFM limitation would be required. This is because Boeing cannot certify a new software configuration with a non-compliance that impacts ability to perform certified system functions. This 737 MAX AFM limitation prohibits RNP AR operations with minima below 0.3 nm for airplanes with the U14.1 FMC OPS and the “C1” FMC hardware installed in FMC-L or FMC-R positions. For 737-7 aircraft, this AFM limitation will apply to airplanes with “C1” FMC hardware regardless of the FMC software installed. At this time, there is no intention to apply the AFM limitation to 737 MAX aircraft (except the 737-7) that have earlier software versions (i.e. U14 FMC OPS and prior), regardless of FMC hardware installed.

Final Action

Boeing Service Bulletin 737-34-3830 has been released for instructions to upgrade to the new "C2" FMC Hardware (P/N: 10-62225-005, GE Model No.: 2907C2, GE P/N: 176200-02-01) due to the change of part number and to restore compliance. A GE Component Service Bulletin is released to modify existing "C1" FMC hardware to the "C2" configuration.

Concurrent Requirements for MAX retrofit:

In order to satisfy Cyber security certification requirement, ONS (Onboard Network System) version OS 9.1 will need to be installed on 737 MAXs in order for C2 to be retrofit

- Reference SB 737-46-1029, FTD 737MAX-FTD-46-22004 for more information on ONS OS 9.1

Milestones

Note: Target dates are based upon currently available status and have risk of future revision. Due to an Issue found in SSA data driving the need for a new, unexpected SSA revision. Recently, the 737 MAX System Safety Assessment (SSA) approval was received on April 7, 2026 and the FAA has approved the SB for the 737 MAX. Latest Schedule below:

737 MAX - FAA AFM Limitation Release | Apr 2022 (Complete)

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737 MAX - EASA & TCCA AFM Limitation Release | May 2022 (Complete)

737 MAX - FMC OPS U14.1 - Boeing Service Bulletin Release | 28-May-2022 (Complete)

737 MAX - FMC Model 2907C2 - FAA Hardware Certification | 1-May-2026 (Complete)

737 MAX - Boeing & GE - Hardware Retrofit Availability | 13-May-2026 (Complete)

737 MAX - FMC SRAM - GE Component Service Bulletin | 13-May-2026 (Complete)

737 MAX - FMC SRAM - Boeing (FAA) Service Bulletin 737-34-3830 | 13-May-2026 (Complete)

737 MAX - FMC Model 2907C2 - EASA Hardware Certification | 17-Jun-2026 (Complete)

737 MAX - FMC SRAM - Boeing (EASA) Service Bulletin | Oct 2026 (Target)

737 MAX - FMC Model 2907C2 Production Incorporation | Sep 2026 (Target)

Operator Action

Boeing encourages operators accomplish noted SB installing the updated FMC hardware and ONS system updates.

Part Information

GE Model 2907C1 FMC (Boeing P/N 10-62225-004, GE P/N 176200-01-01) is the subject of the non-compliance.

A limited seed pool will be available to assist operators to cycle upgrades through their fleet. GE is available to assist with planning upgrade schedules and allocation of seed units.

For assistance with MRO or operator-performed service bulletin upgrades, please contact your GE representative: terry.l.moorman@ge.com

Interchangeability:

GE FMC Model No. 2907C1 (Boeing P/N: 176200-01-01) and GE FMC Model No. 2907C2 (Boeing P/N: 176200-02-01) are **not interchangeable** and **not intermixable**.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Bulletin (SB)	737-34-3830		
Service Bulletin (SB)	737-46-1029		
Production Revision Record (PRR)	3M0290		
Fleet Team Digest Article	737MAX-FTD-46-22004		

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(FTD)			

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Model C2 FMC	10-62225-005	GE Aviation	176200-02-01
SRAM		GE Aviation	173210-11
SRAM		GE Aviation	173210-13
Model C1 FMC	10-62225-004	GE Aviation	176200-01-01

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Flight Operations : Flight Operations
- New Equipment : Information on new parts available now or soon to be introduced

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